HAPPY

ACADEMIC DETAILS

Year	Degree / Board	Institute	GPA / Marks(%)
---	M.Tech in Computer Science & Engineering	Indian Institute of Technology Delhi	7.95
2024	B.Tech in Artificial Intelligence and Data Science	IIITDM Kurnool	8.42
2020	High School / CBSE	JP DAV Public School, Ganaur	92.80
2018	Intermediate / CBSE	JP DAV Public School, Ganaur	87.00

IIT DELHI THESIS

Title: Representation Learning using ECG+PPG
Supervisor: Prof. Rahul Garg
Description: Developing a **multi-modal representation learning** framework to fuse **ECG and PPG** signals, creating a robust feature space to mitigate noise and artifacts in PPG data. This research focuses on developing a novel deep learning model that combines ECG and PPG signals to improve diagnostic accuracy. The project included clinical validation with data from **AIIMS Hospital**, demonstrating practical application.

INTERNSHIPS

- **CSIR(Council of Scientific & Industrial Research) Research Internship , Chennai** (June, 2023 - July, 2023)
  - Developed a highly efficient **GAN model** to deal with the problem of limited training images which ultimately helps in developing a highly accurate Classification and Segmentation model with accuracy 1.00 in classification and IoU 0.70 in segmentation.
- **Samsung Prism Project Internship , Samsung R&D Institute (Virtual)** (Dec, 2022 - June, 2023)
  - Project involves the integration of UNet and GAN models, incorporating **Semi-supervision** and **Active Learning** techniques which improved **semantic segmentation** model accuracy from 0.66 to 0.73.
  - Successfully managed 3 complex datasets (Cityscapes, DLRSD, Minifrance) and collaborated closely with SamsungR&D group, aligning AI model outcomes with practical needs, showcasing adaptability and teamwork.

PROJECTS

- **Stroke Type Detection using Fundus Imaging (Minor Project) (Prof. Rahul Garg)**:
  - Developed a deep learning model for stroke type detection using fundus imaging, fine-tuning the **RETFound foundation model** in a collaborative research project with AIIMS Delhi,
  - It focuses on applying computer vision to medical diagnostics. The project also involved multi-class classification for Diabetic Retinopathy and stroke detection.
- **AI Tool for Semantic Segmentation (Dr. Shounak Chakraborty , IIIT Kurnool)** :
  - Designed and implemented a **semi-supervised learning** pipeline that leveraged both labeled and unlabeled data, achieving a performance improvement from 0.66 to 0.73 without extensive manual annotation. Developed a user-friendly Tkinter GUI for intuitive visualization.
  - Integrated the tool with drone technology to enable real-time in-flight semantic segmentation.
- **Visual Question Answering using Multi-modal Transformers (Prof. Parag Singla)** :
  - Developed a Visual Question Answering (VQA) model by implementing a multi-modal transformer architecture in PyTorch using CLEVR Dataset.
  - Engineered a **cross-attention** feature fusion mechanism to integrate visual and textual features, creating a joint representation for accurate prediction. The model was trained end-to-end and fine-tuned to enhance performance.
- **Object Detection with DETR (Prof. Chetan Arora)** :
  - Evaluated and fine-tuned a pre-trained Deformable DETR model for object detection on the Foggy Cityscapes dataset.
  - Conducted comprehensive quantitative and qualitative analysis of model performance. Generated precision-recall tables and computed **mean Average Precision (mAP)** to evaluate the pre-trained and fine-tuned models.

TECHNICAL SKILLS

- **Languages:** C/C++, Python, Javascript, HTML, CSS, Flex, Bison, SQL
- **Libraries / Tools:** Tensorflow, Pytorch, Tkinter, Swing API(Java), Qt, VScode, Git, Github, Colab, Verilog, Unity

SCHOLASTIC ACHIEVEMENTS

- **Departmental Rank 6:** Current rank in M.Tech (Computer Science & Engineering), IIT Delhi.
- **Rank 1 at Facet Interface 2022(Bits Pilani):** Ranked 1st out of 450 teams in Machine Learning competition.
- **Rank 1 in National Abacus Championship:** Got Rank-1 out of 2000 participants in withAbacus category.

QUALIFYING EXAMS

- **Graduate Aptitude Test in Engineering (GATE) Rank:** 29

HAPPY

IIT COURSE

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M.Tech in Computer Science & Engineering	Indian Institute of Technology Delhi	7.95	---

COURSES DONE

Advanced Data Structures, Intro. To Logic & Funct. Prog., Software Systems Laboratory, Teaching Practicum In Computer Science And Engineering, Computer Vision, Machine Learning, Minor Project, Teaching Practicum In Computer Science And Engineering